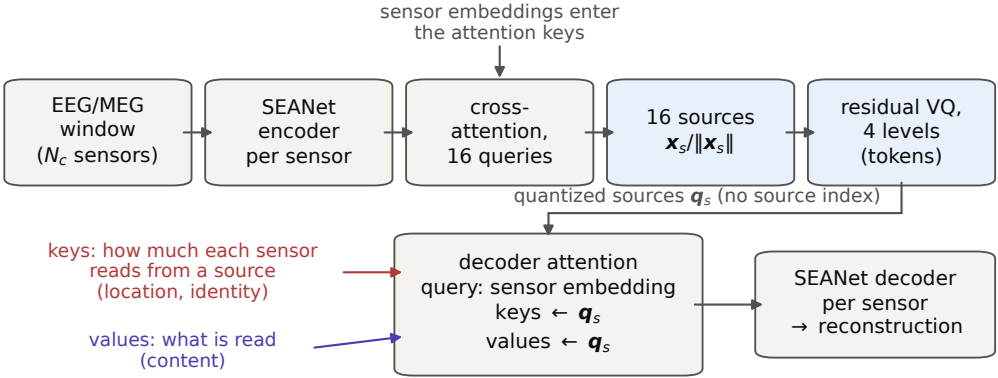
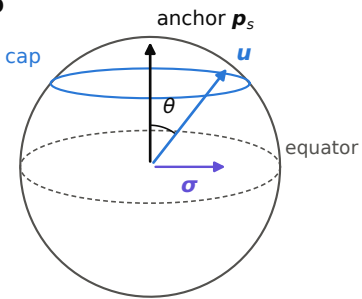
a



b



$$u = a\,p_s + \sqrt{1-a^2}\,\sigma$$

p_s: anchor (normalized mean of the vectors)

θ: colatitude (angle to the anchor)

$a = \cos\theta$: anchor coordinate

$\sigma \perp p_s$: tangent direction (unit vector)